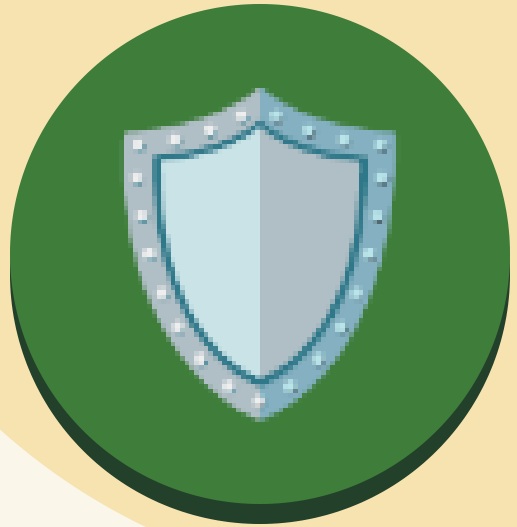
Small Engines Overhaul · Unit 0 · Level 2 (modified)

Shop Start-Up & Safety



TODAY'S PLAN

- 📝 Syllabus and contract
- 🥽 Personal protective equipment
- ⛽ Fire and fuel safety
- 🧰 Clean-up and norms

Review · 5 minutes

Quick Review



Name two machines that use a small gas engine.

What do you already know about shop safety?

Where is the shop exit?

Words to Know

Words to Know



Safety glasses

Clear glasses that protect your eyes from flying parts and fluids. Worn the whole time in the shop.



Coveralls

One-piece work clothes that keep oil and grease off your body and clothes.



Closed-toe shoes

Sturdy shoes that protect your feet from dropped parts and tools.



Flammable

Able to catch fire easily, like gasoline and oily rags.

Words to Know

More Words to Know



🧯 **Fire extinguisher**: A red tank that sprays to put out small fires. Know where it is.

🏭 **Carbon monoxide**: A deadly gas in engine exhaust that you cannot see or smell.

🧰 **Tool board**: The wall or cart where every tool has its own spot.

📝 **Contract**: A paper you and your parent sign that says you will follow the shop rules.

Learn

Personal Protective Equipment (PPE)



Safety glasses
whole time in the shop



Coveralls
keep oil off you



Closed-toe shoes
protect feet



No loose items
tie hair, no dangling jewelry

🃏 Check 1 of 3



When do you wear safety glasses in the shop?

Hold up 1, 2, or 3:

1️⃣ Only when grinding

2️⃣ The whole time

3️⃣ Only when the teacher looks

✅ Answer · Card 1 of 3



The whole time

Parts and fluids can fly at any time.

Learn

Tools and Supervision



All shop tools are used **with adult supervision**.

Watch the demo, then try it while an adult watches.

Use the right tool for the job; report broken tools.

Learn

Fuel and Fire



Gasoline is **flammable**; its fumes can ignite from a spark.

Oily rags go in the **metal safety can**.

Know where the **fire extinguisher** and exits are.

🃏 Check 2 of 3



Where do oily rags go?

Hold up 1, 2, or 3:

1️⃣ In the regular trash

2️⃣ In your pocket

3️⃣ In the metal safety can

✅ Answer · Card 2 of 3



In the metal safety can

Oily rags can catch fire on their own.

Learn

Exhaust and Hot Parts



Exhaust has **carbon monoxide**, a gas you cannot see or smell.

Engines run only outdoors or with ventilation, with an adult.

Mufflers and cylinders stay hot; let them cool.

Learn

Before You Work on an Engine

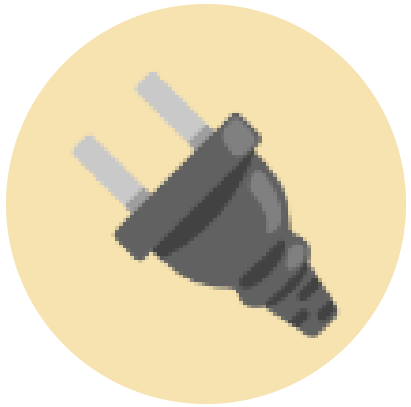


Disconnect the **spark plug wire** so it cannot start.

Drain or shut off fuel with adult supervision.

Lift heavy engines with a partner.

🃏 Check 3 of 3



Why disconnect the spark plug wire first?

Hold up 1, 2, or 3:

1️⃣ So the engine cannot start

2️⃣ To make it lighter

3️⃣ To save gas

✅ Answer · Card 3 of 3



So the engine cannot start

Turning the blade or flywheel could start it.

Steps

Syllabus Rules to Get Into the Shop



1. Sign the syllabus and shop contract with your parent
2. Finish the class assignments
3. Pass the safety test (80%)
4. Then you get an engine

Wrap-Up

Recap



PPE
glasses, coveralls, shoes



Fuel
flammable, safety can



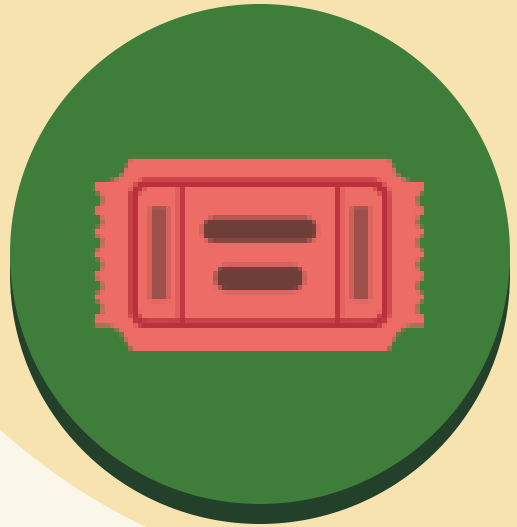
Spark plug wire
off before work



80%
pass to enter the shop

Wrap-Up

Exit Ticket



List the 3 pieces of PPE.

Name 2 fuel safety rules.

What must you pass before working in the shop?